TP Nº 2 – PROBLEMA P11

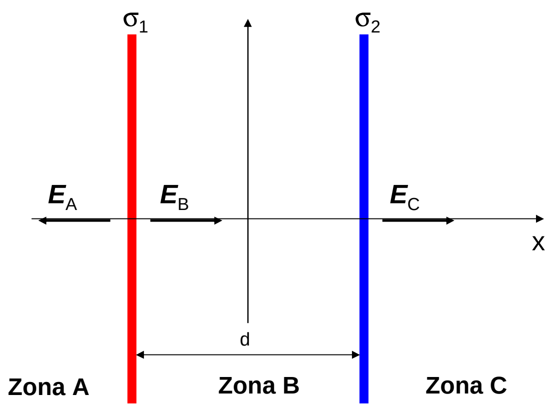
- Considere dos planos no conductores paralelos infinitamente extendidos y separados una distancia D. Sobre cada uno de ellos yacen distribuciones uniformes de cargas de densidades superficiales σ_1 y σ_2, respectivamente. Determine el campo electrostático en todas partes y en cada uno de los siguientes casos:

a) $\sigma_1 = 3\ mC/m^2$ y $\sigma_2 = 3\ mC/m^2$
b) $\sigma_1 = 3\ mC/m^2$ y $\sigma_2 = -3\ mC/m^2$
c) $\sigma_1 = 3\ mC/m^2$ y $\sigma_2 = 5\ mC/m^2$
d) $\sigma_1 = 3\ mC/m^2$ y $\sigma_2 = -5\ mC/m^2$

E.E.Grumel - 2020

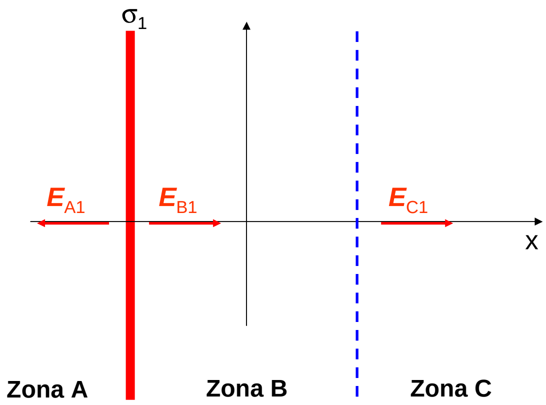
Solución:

Hallaremos la solución de este problema en forma general.

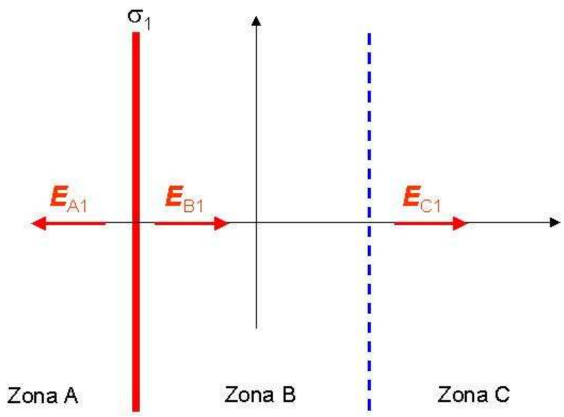


1er. Paso: Hallamos el campo eléctrico que genera la placa 1 en todas las zonas.

Para la placa 1 hallamos:

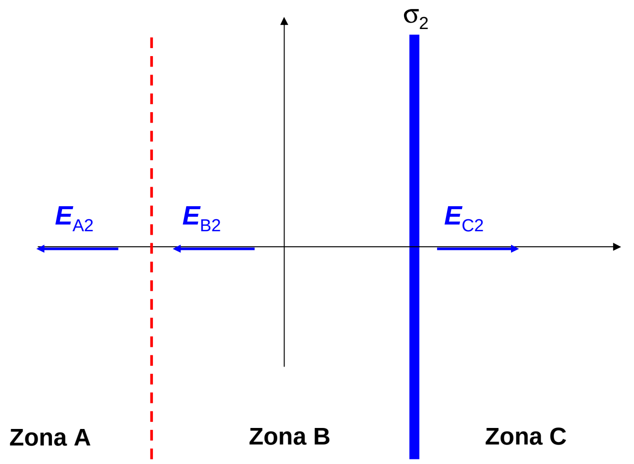


$$\vec{E}_{A1} = \frac{\sigma_1}{2\varepsilon_0}\left(-\hat{i}\right)$$

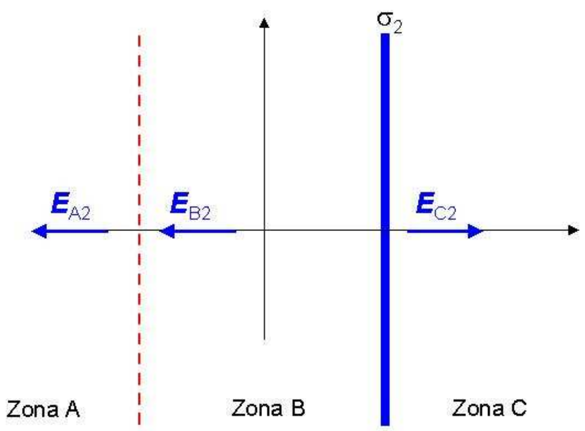


$$\vec{E}_{B1} = \vec{E}_{C1} = \frac{\sigma_1}{2\varepsilon_0}\left(\hat{i}\right)$$

Para la placa 2:

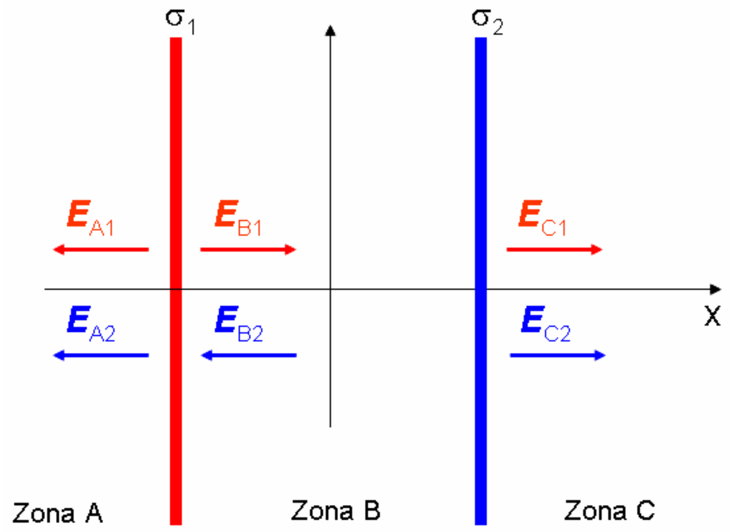


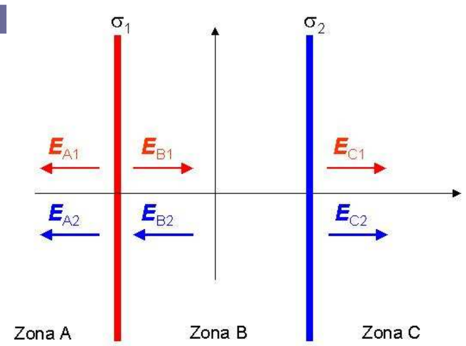
$$\vec{E}_{A2} = \vec{E}_{B2} = \frac{\sigma_2}{2\varepsilon_0}\left(-\hat{i}\right)$$



$$\vec{E}_{C2} = \frac{\sigma_2}{2\varepsilon_0}\left(\hat{i}\right)$$

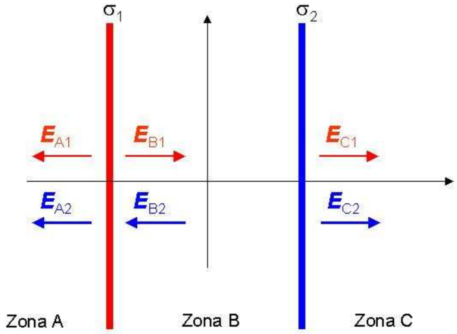
2do. Paso: Aplicamos el ppio de superposición:





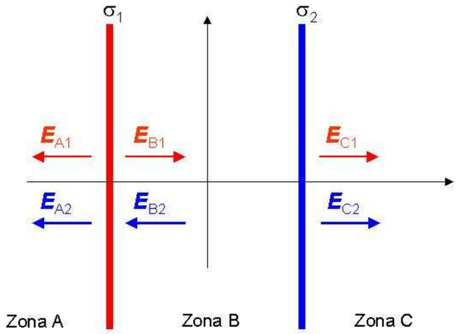
$$\vec{E}_A = \vec{E}_{A1} + \vec{E}_{A2} = \frac{\sigma_1}{2\varepsilon_0}\left(-\hat{i}\right) + \frac{\sigma_2}{2\varepsilon_0}\left(-\hat{i}\right)$$

$$\vec{E}_A = \frac{(\sigma_1 + \sigma_2)}{2\varepsilon_0}\left(-\hat{i}\right)$$



$$\vec{E}_B = \vec{E}_{B1} + \vec{E}_{B2} = \frac{\sigma_1}{2\varepsilon_0}\left(\hat{i}\right) + \frac{\sigma_2}{2\varepsilon_0}\left(-\hat{i}\right)$$

$$\vec{E}_B = \frac{\left(\sigma_1 - \sigma_2\right)}{2\varepsilon_0}\left(\hat{i}\right)$$



$$\vec{E}_C = \vec{E}_{C1} + \vec{E}_{C2} = \frac{\sigma_1}{2\varepsilon_0}\left(\hat{i}\right) + \frac{\sigma_2}{2\varepsilon_0}\left(\hat{i}\right)$$

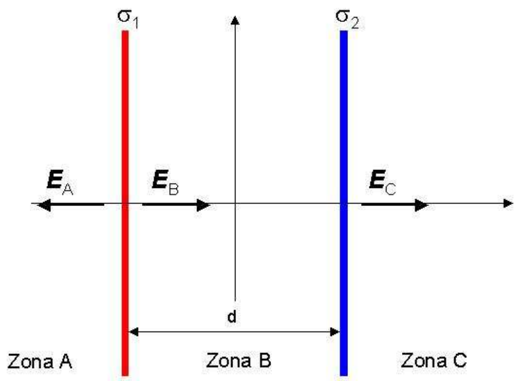
$$\vec{E}_C = \frac{\left(\sigma_1 + \sigma_2\right)}{2\varepsilon_0}\left(\hat{i}\right)$$

Resumiendo:

$$\vec{E}_A = \frac{(\sigma_1 + \sigma_2)}{2\varepsilon_0}(-\hat{i})$$

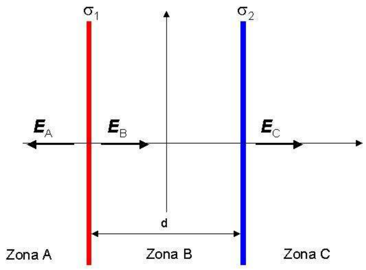
$$\vec{E}_B = \frac{(\sigma_1 - \sigma_2)}{2\varepsilon_0}(\hat{i})$$

$$\vec{E}_C = \frac{(\sigma_1 + \sigma_2)}{2\varepsilon_0}(\hat{i})$$



Casos límites:

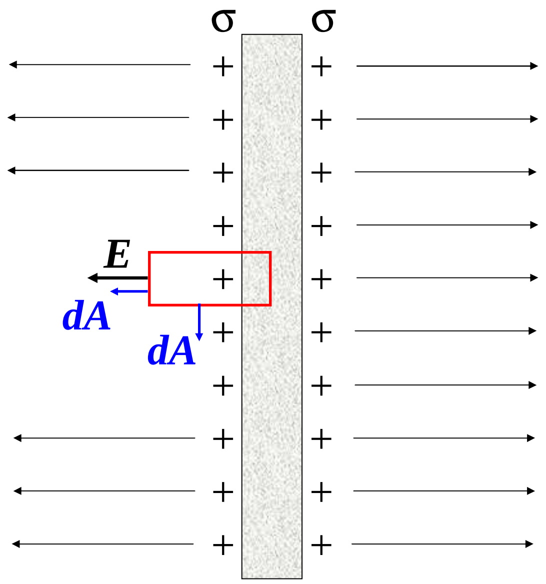
a) $\sigma_1 = \sigma_2 = \sigma$



$$\vec{E}_A = \frac{(\sigma + \sigma)}{2\varepsilon_0}(-\hat{i}) = \frac{\sigma}{\varepsilon_0}(-\hat{i})$$

$$\vec{E}_B = \frac{(\sigma - \sigma)}{2\varepsilon_0}(\hat{i}) = 0$$

$$\vec{E}_C = \frac{(\sigma + \sigma)}{2\varepsilon_0}(\hat{i}) = \frac{\sigma}{\varepsilon_0}(\hat{i})$$



$E_{cond} = 0$

Casos límites:

b) $\sigma_1 = -\sigma_2 = \sigma$

$$\vec{E}_A = \frac{(\sigma - \sigma)}{2\varepsilon_0}(-\hat{i}) = 0$$

$$\vec{E}_B = \frac{(\sigma - (-\sigma))}{2\varepsilon_0}(\hat{i}) = \frac{\sigma}{\varepsilon_0}(\hat{i})$$

$$\vec{E}_C = \frac{(\sigma - \sigma)}{2\varepsilon_0}(\hat{i}) = 0$$